

For teacher use only. Full worked answers are shown. MC answer summary: B C C C D B B C B C.

Part I — Multiple Choice

- (B) The distribution of all possible values of a statistic over all possible samples. That is the definition of a sampling distribution.
- (B) $z = (75 - 69.2)/3.6 \approx 1.61$; $P(Z > 1.61) \approx 0.0537$.
- (C) $\sigma_{\bar{x}} = \sigma/\sqrt{n} = 10/\sqrt{25} = 10/5 = 2$.
- (C) The CLT requires $n \geq 30$ (or a normal population). Option (A) is sufficient but not necessary; (D) is the Large Counts condition for proportions.
- (D) The sample maximum is biased high — it consistently overestimates the population maximum.
- (B) $\sigma_{\hat{p}} = \sqrt{0.40(0.60)/100} = \sqrt{0.0024} \approx 0.0490$.
- (B) $\sigma_{\bar{x}} = 16/\sqrt{64} = 2$; $z = (122 - 120)/2 = 1.00$; $P(\bar{x} > 122) = 1 - 0.8413 = 0.1587$.
- (C) The Large Counts condition: $np \geq 10$ and $n(1 - p) \geq 10$.
- (B) $\mu_{\bar{x}_1 - \bar{x}_2} = \mu_1 - \mu_2 = 80 - 75 = 5$.
- (C) Larger n reduces $\sigma_{\bar{x}} = \sigma/\sqrt{n}$, narrowing the sampling distribution around the true parameter.

Part II — Free Response

- Exercise habits, $p = 0.28$, $n = 200$.
 - (1) Random: stated — the sample is a random sample. ✓
 (2) 10% condition: $n = 200 \leq 10\%$ of all adults (millions). ✓
 (3) Large Counts: $np = 200(0.28) = 56 \geq 10$; $n(1 - p) = 200(0.72) = 144 \geq 10$. ✓
 All conditions met; use the normal model for $\hat{p}$.
 - $\mu_{\hat{p}} = 0.28$; $\sigma_{\hat{p}} = \sqrt{0.28(0.72)/200} = \sqrt{0.001008} \approx 0.03175$.
 - $z = (0.32 - 0.28)/0.03175 \approx 1.260$. $P(\hat{p} > 0.32) = P(Z > 1.260) \approx 1 - 0.8962 = 0.1038$. About 10.4%.
 - If the true proportion of adults who exercise at least 5 days per week is 28%, there is approximately a 10.4% chance that a random sample of 200 adults would yield a sample proportion of more than 32%.

Teacher Note

Award partial credit on (c) if the student correctly standardizes but makes a minor arithmetic error. Require a sentence with context for full credit on (d).

- Bolt lengths $N(50, 1.5)$.
 - $z = (51.5 - 50)/1.5 = 1.00$; $P(X > 51.5) = 1 - 0.8413 = 0.1587$. About 15.9%.
 - (1) Random: stated. ✓
 (2) 10%: $36 \leq 10\%$ of all bolts produced. ✓
 (3) Normal/Large Sample: population is approximately normal. ✓
 All conditions met.

- (c) $\mu_{\bar{x}} = 50 \text{ mm}$; $\sigma_{\bar{x}} = 1.5/\sqrt{36} = 0.25 \text{ mm}$.
- (d) $z = (50.5 - 50)/0.25 = 2.00$; $P(\bar{x} > 50.5) = 1 - 0.9772 = 0.0228$. **About 2.3%.**
- (e) **The probability for a single bolt (15.9%) is much larger than for the sample mean (2.3%). Averaging 36 bolts cancels out individual extremes: the standard error ($\sigma_{\bar{x}} = 0.25$) is much smaller than the population SD ($\sigma = 1.5$), so sample means are far less variable than individual measurements.**
- (f) $z = (49.4 - 50)/0.25 = -2.40$; $P(\bar{x} < 49.4) = P(Z < -2.40) \approx 0.0082$. **About 0.82%. This is unusual (probability < 5%); such a sample mean would occur by chance less than 1% of the time and warrants investigation.**

13. Homework completion, two schools.

- (a) $n_1 p_1 = 120(0.65) = 78 \geq 10$. ✓
 $n_1(1 - p_1) = 120(0.35) = 42 \geq 10$. ✓
 $n_2 p_2 = 150(0.58) = 87 \geq 10$. ✓
 $n_2(1 - p_2) = 150(0.42) = 63 \geq 10$. ✓
All four Large Counts conditions met.
- (b) $\mu_{\hat{p}_1 - \hat{p}_2} = 0.65 - 0.58 = 0.07$.
 $\sigma = \sqrt{0.65(0.35)/120 + 0.58(0.42)/150} = \sqrt{0.001896 + 0.001624} = \sqrt{0.003520} \approx 0.05933$.
- (c) $z = (0.15 - 0.07)/0.05933 \approx 1.348$; $P(\hat{p}_1 - \hat{p}_2 > 0.15) \approx 1 - 0.9112 = 0.0888 \approx 8.9\%$.

MC Answer Key: 1-B 2-B 3-C 4-C 5-D 6-B 7-B 8-C 9-B 10-C

Teacher Note

Scoring guidance. Free response is scored holistically: award points for correct conditions checked, correct formulas stated, correct calculations shown, and contextual interpretations. A correct answer with no work earns at most 50% on calculation steps. Context is required in interpretations — deduct 1 point if the student gives a correct probability statement but omits reference to the real-world scenario.